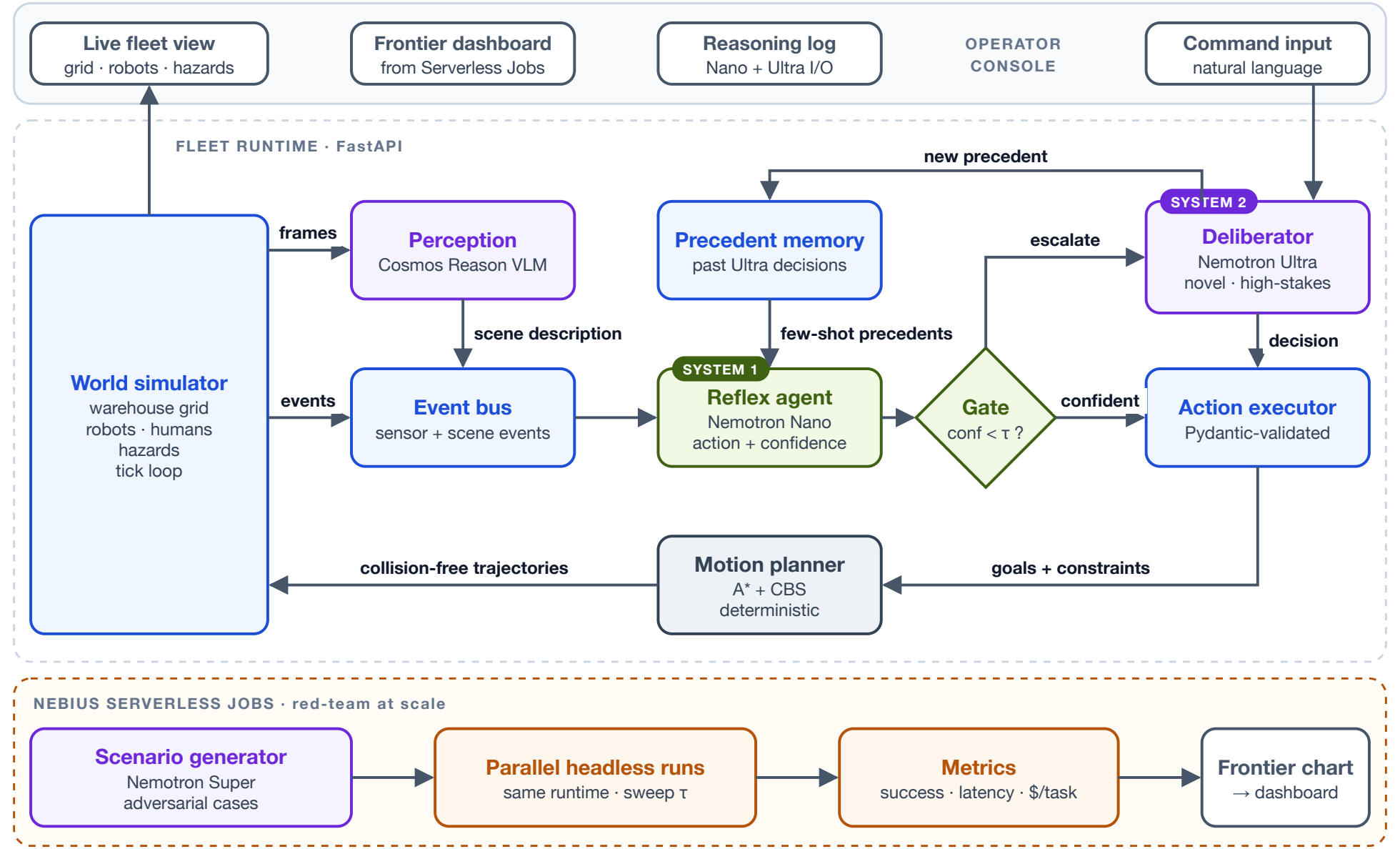
NemoFleet · System 1 / System 2 for robot fleets

SYSTEM MAP

A fast model on each robot handles routine events; a large reasoning model in the cloud handles only what is novel or high-stakes. A deterministic planner owns every trajectory, so paths are collision-free by construction.

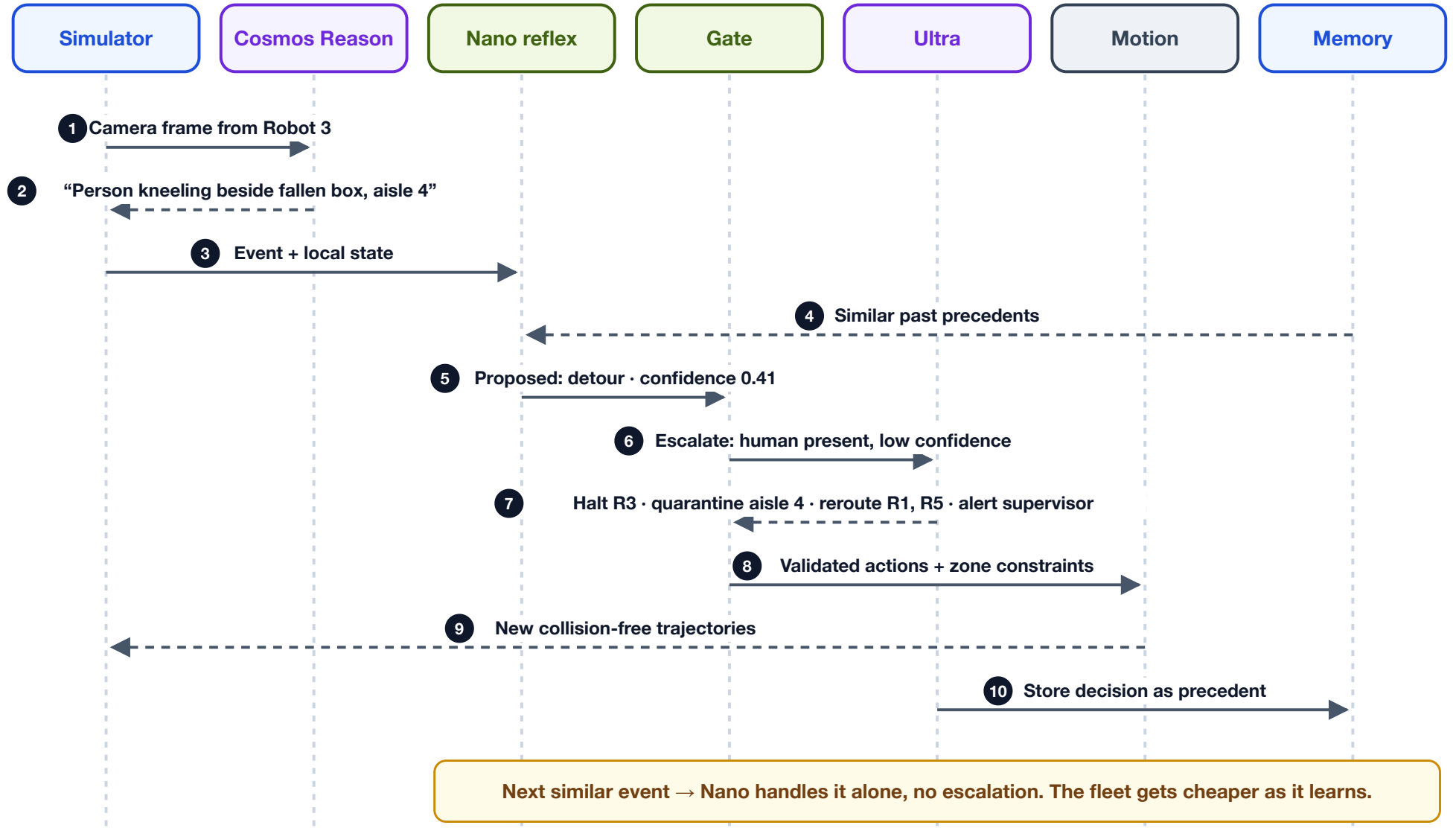


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A person enters a robot lane

ESCALATION SEQUENCE

Cosmos turns pixels into meaning, Nano proposes a response with low confidence, the gate escalates, Ultra decides, the planner reroutes, and the decision becomes a precedent for next time.



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- - - dashed arrow = response / return

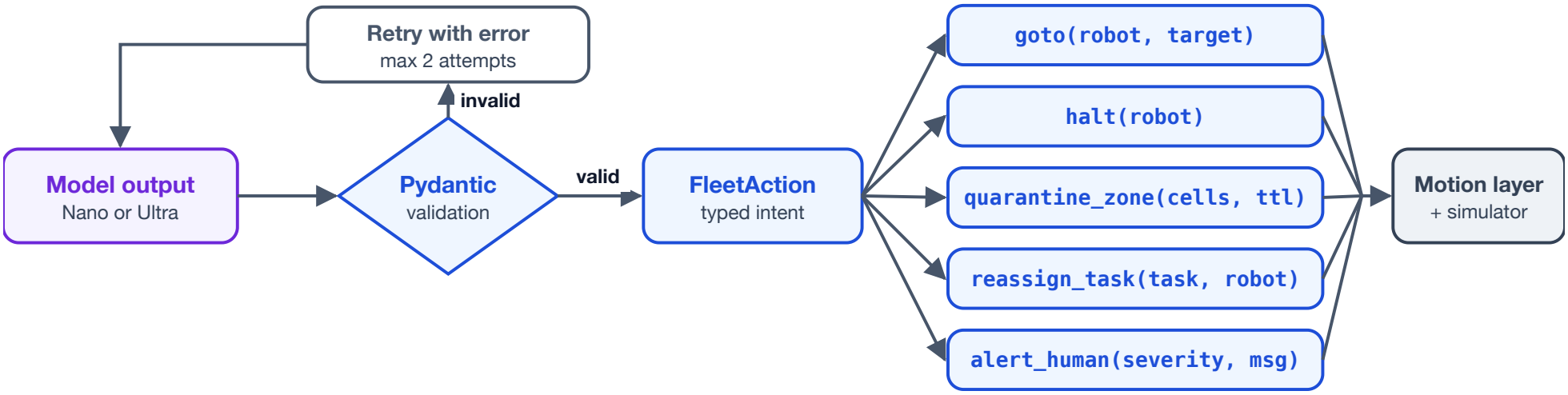
Who decides what

RESPONSIBILITIES AND SAFETY BOUNDARY

> **Principle:** models decide *what* should happen; the planner decides *how* to move. Every model output must pass schema validation before anything moves.

LAYER	MODEL / ALGORITHM	DECIDES	LATENCY	NEVER DOES
Perception	Cosmos Reason	What the camera sees	~1 s	Plan actions
Reflex	Nemotron Nano	Routine responses + a confidence score	< 1 s	Handle safety-critical novel events alone
Escalation gate	Rules + threshold τ	Whether to call System 2	~0	Generate content
Deliberation	Nemotron Ultra	Novel, high-stakes decisions; task re-prioritization	5–30 s	Produce raw coordinates
Motion	A* + CBS	Exact collision-free paths	< 10 ms	Interpret semantics
Executor	Pydantic schemas	Whether an action is valid	~0	Accept malformed LLM output

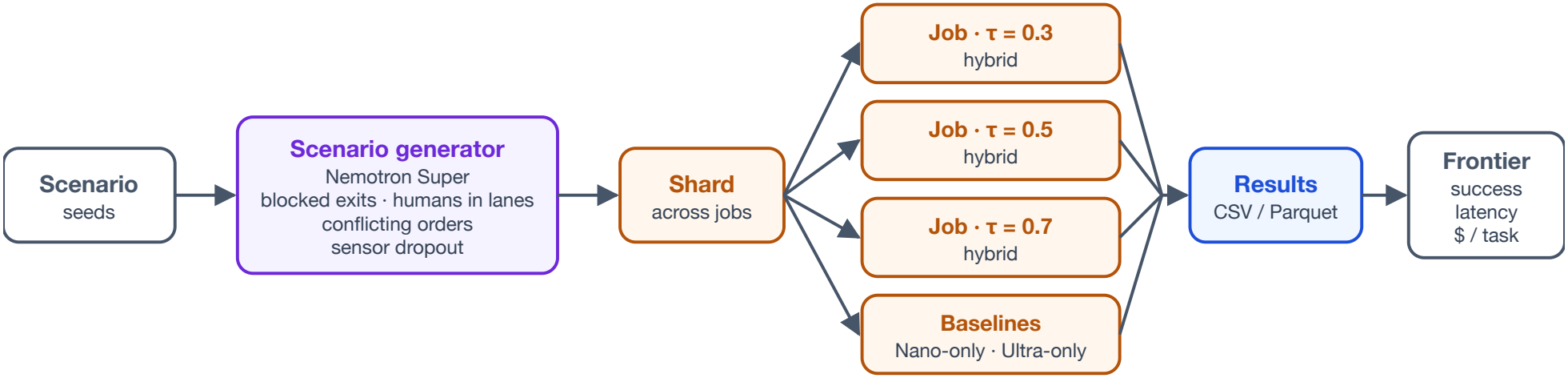
Action schema boundary



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Measuring when to escalate

Nemotron generates adversarial warehouse scenarios, Serverless Jobs run them in parallel at several escalation thresholds τ, and the results map the success / latency / cost frontier of the hybrid against Nano-only and Ultra-only baselines.



Where NVIDIA and Nebius are used

COMPONENT	NVIDIA MODEL	NEBIUS SERVICE
Reflex agent	Nemotron Nano	Token Factory
Deliberator	Nemotron Ultra	Token Factory
Perception	Cosmos Reason	Token Factory or AI Cloud endpoint
Scenario generator	Nemotron Super	Token Factory
Red-team runs	—	Serverless Jobs

Model IDs to be confirmed against the Token Factory catalog.

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Code map: nemofleet/

PLANNED MODULE LAYOUT

models.py

WorldState, Robot, Event
FleetAction, Decision

sim/

world.py — grid, robots, humans, tick loop
scenarios.py — loading + generation

motion/

astar.py — single-agent paths
cbs.py — multi-agent conflict search

cognition/

client.py — Token Factory client
reflex.py — Nano · System 1
deliberator.py — Ultra · System 2
gate.py — escalation policy
perception.py — Cosmos Reason
memory.py — precedent store

api/

server.py — /state /command /inject /tick /log

jobs/

redteam.py — Serverless Job entry
frontier.py — aggregation + charts

ui/

Operator console: live view, command input, reasoning log, frontier dashboard

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